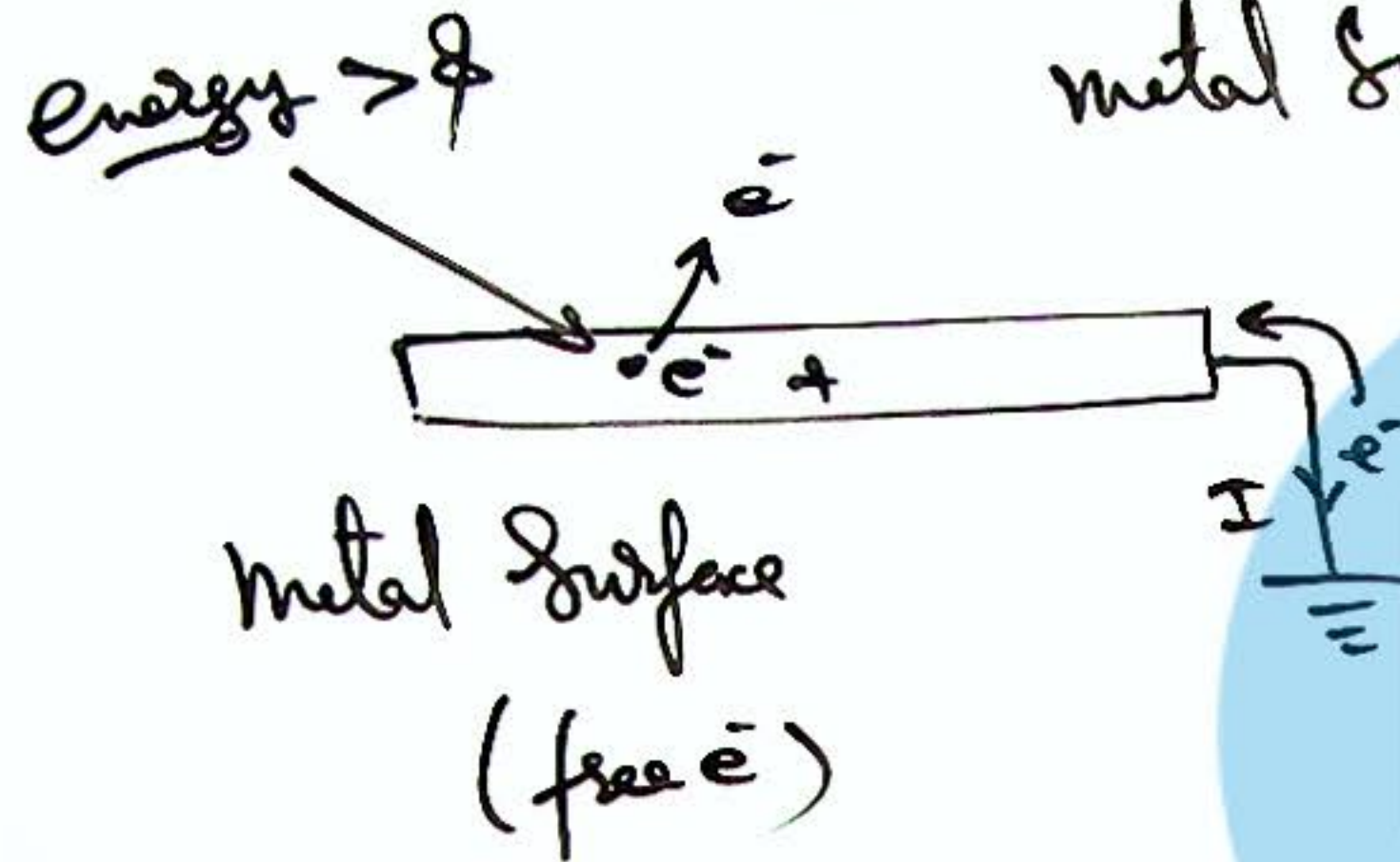


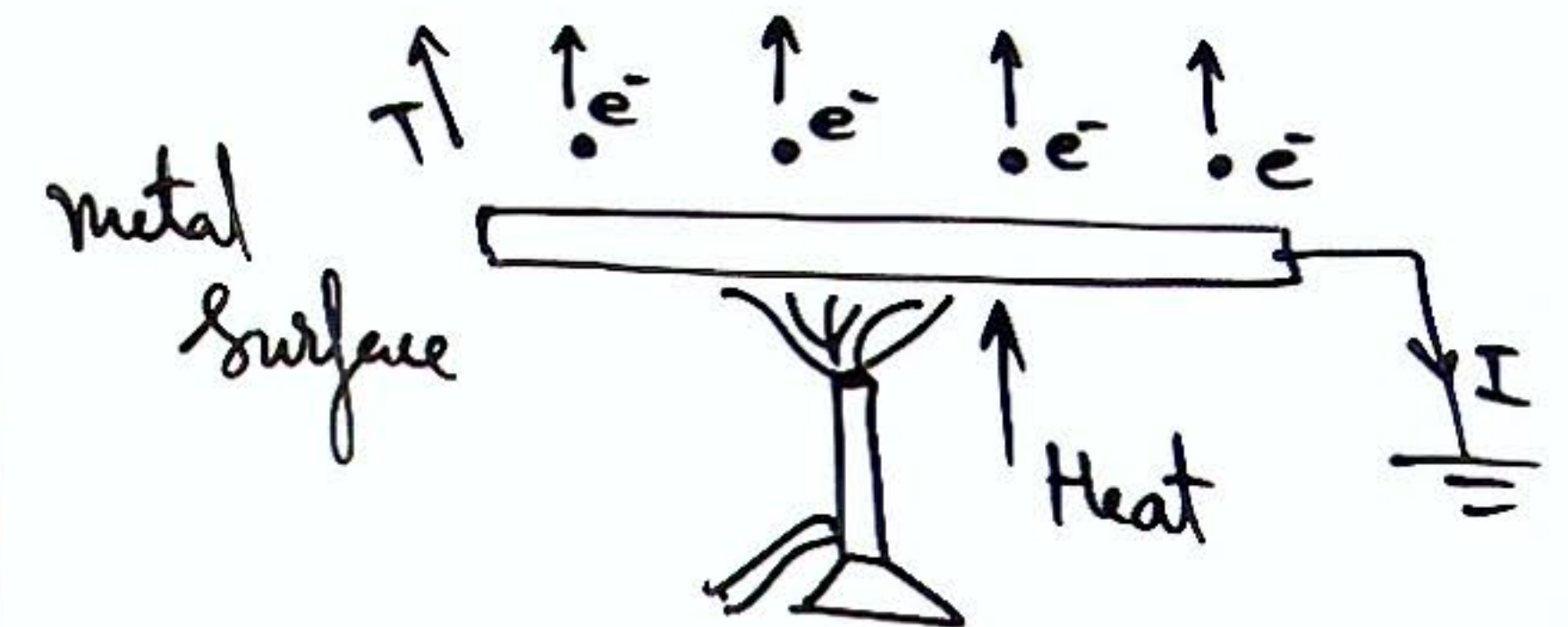
① Electron Emission Processes:

work function $\rightarrow$ The min energy required ' ϕ ' to eject an e^- from a metal surface.

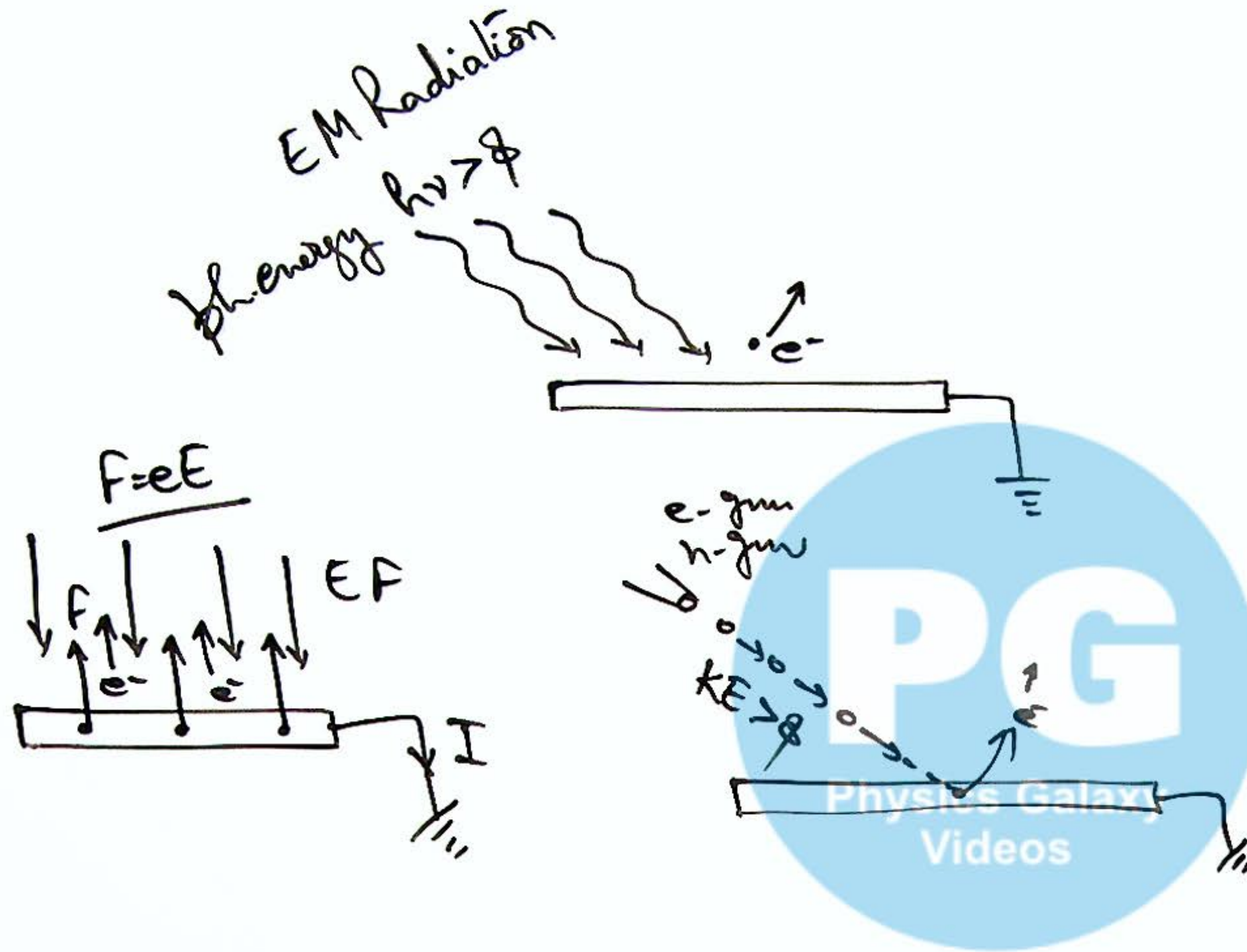


(i) Thermionic Emission

Energy $\rightarrow$ Heat



① Electron Emission Processes:



- (i) Thermionic Emission
- (ii) Photoelectric Emission
Energy $\rightarrow$ EM Radiation
- (iii) Field Emission
Energy $\rightarrow$ by EF
- (iv) Secondary Emission
Energy $\rightarrow$ by collision

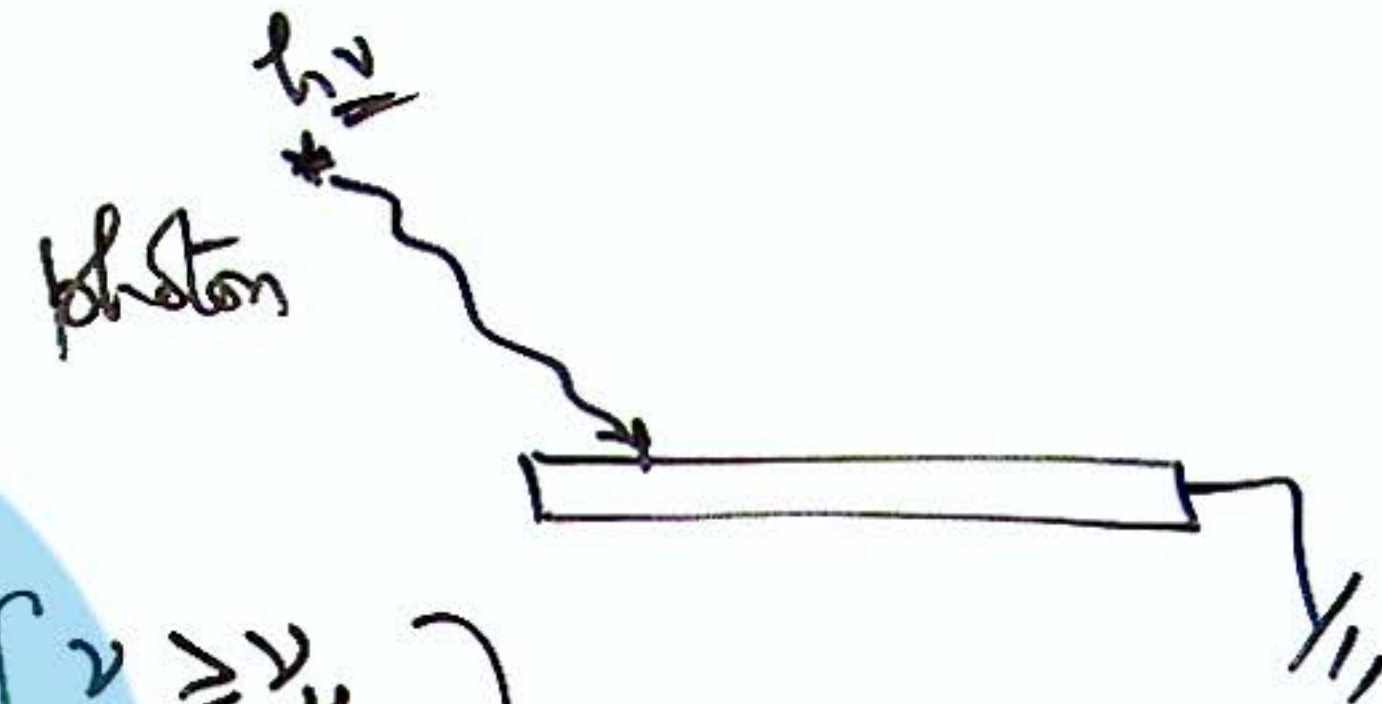
② Threshold Frequency $\rightarrow \nu_{th}$
Threshold Wavelengths $\rightarrow \lambda_{th}$
 $\rightarrow$ min freq to start PEE

Such that

$$h\nu_{th} = \phi$$

$$\frac{hc}{\lambda_{th}} = \phi$$

$$\text{for PEE } \left[\begin{array}{l} \nu \geq \nu_{th} \\ \lambda \leq \lambda_{th} \end{array} \right]$$

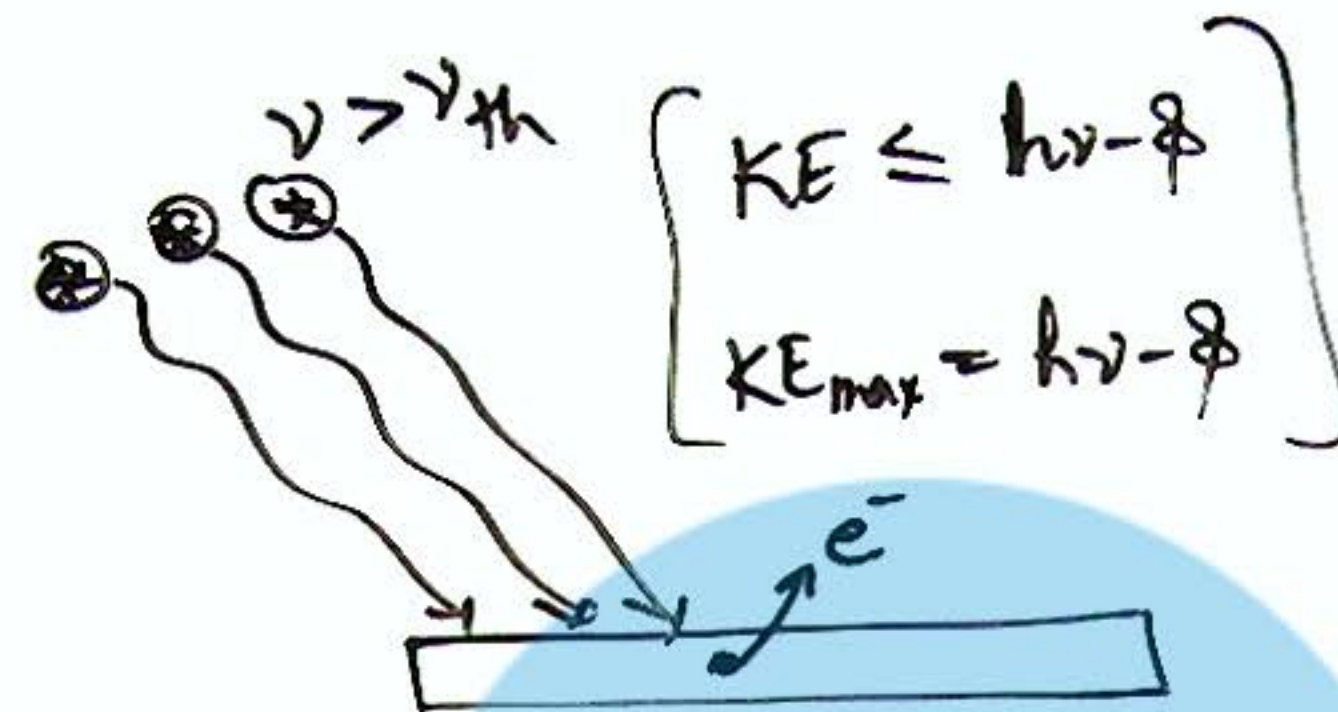


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③ Behaviour of photon energy in PEE:

photon is an
energy quantum
(pkt of energy)

it is either fully absorbed
or not absorbed.



$W/F = \phi$

Incident energy

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$$h\nu = \phi + \frac{1}{2}mv_{max}^2$$

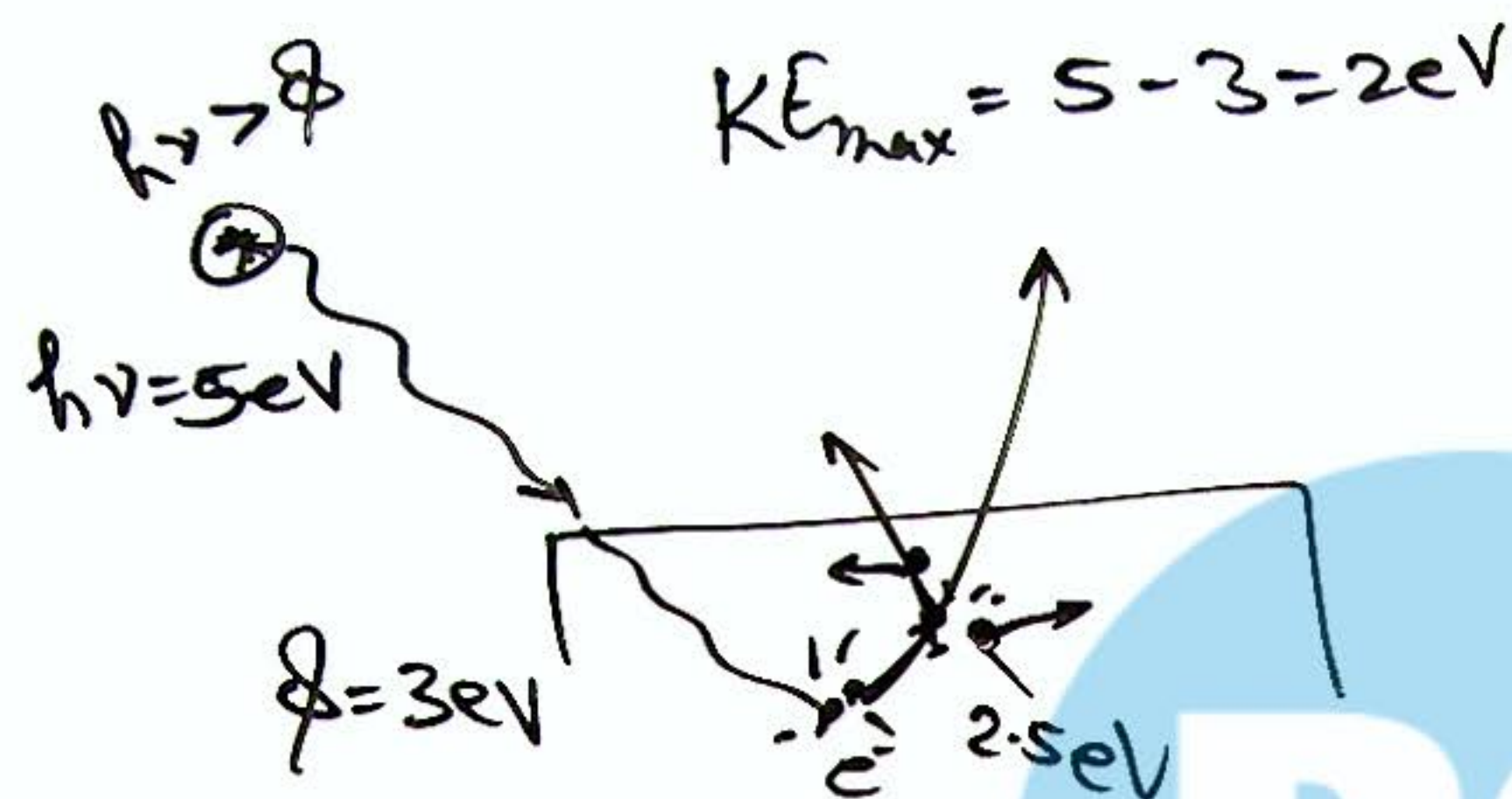
Absorbed photon energy
is used by e^- in 2 parts

① In W/F

② As KE after ejection

④ Quantum Efficiency of a Metal Surface :

' η '



It is not necessary that every photon ($h\nu > \phi$) will eject an e^- .

$$\eta = \frac{\text{no of ejected photoelectrons/s}}{\text{no of total incident photons/s}} \times 100$$

PG

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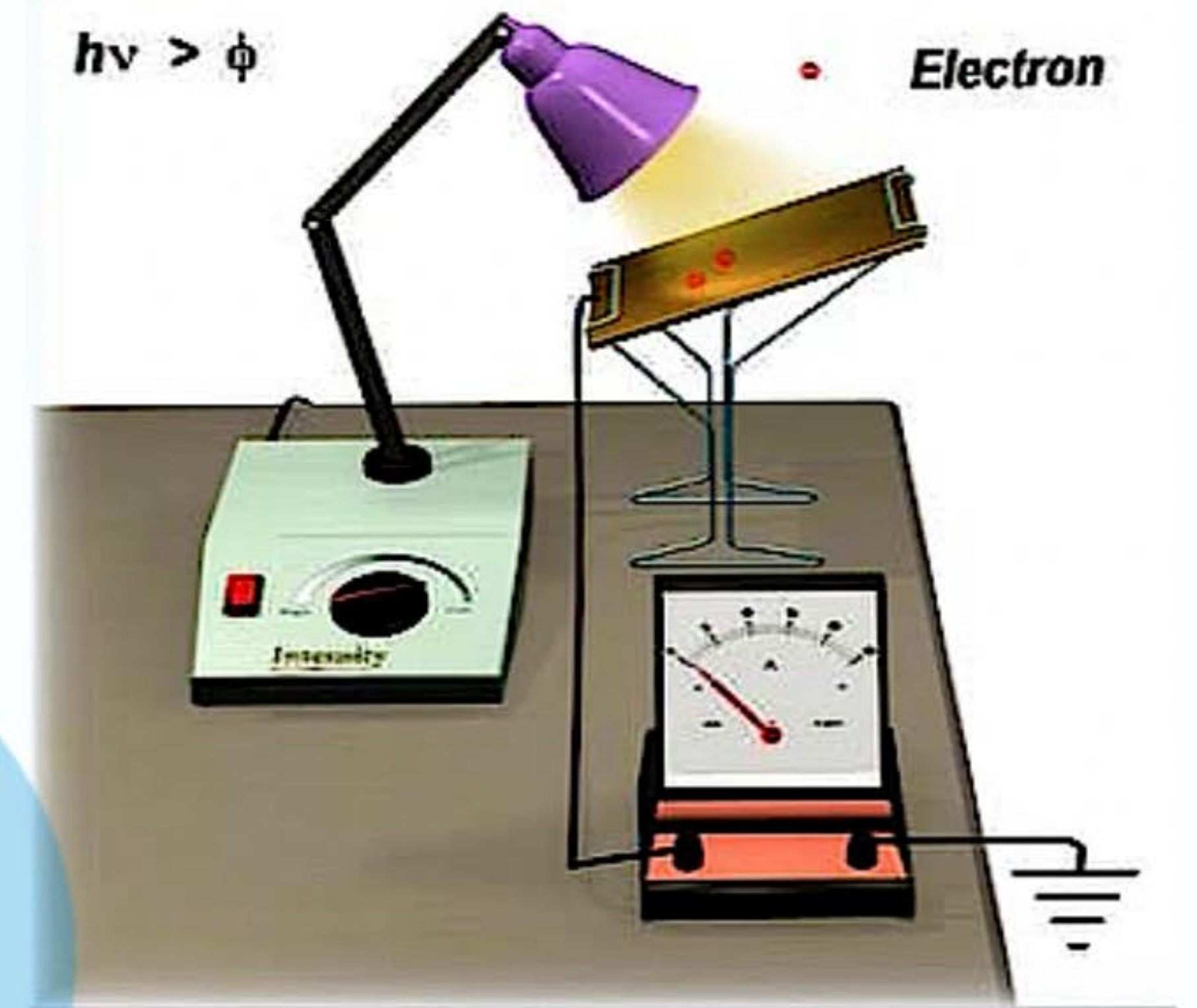
⑤ Effect of Intensity variation in PEE:

Intensity $\propto$ photon flux
no of ph/secm²

As Intensity $\uparrow \Rightarrow$ Current $\uparrow$

$$\underline{KE_{\max} = h\nu - \phi}$$

As freq $\uparrow \Rightarrow$ KE of e^- ejected $\uparrow$



⑥ Effect of Temperature on PEE:

"No effect"

As PEE is a quantum phenomenon.

So it is independent from thermionic emission.

⑦ Time lag of PEE e^- ejection:

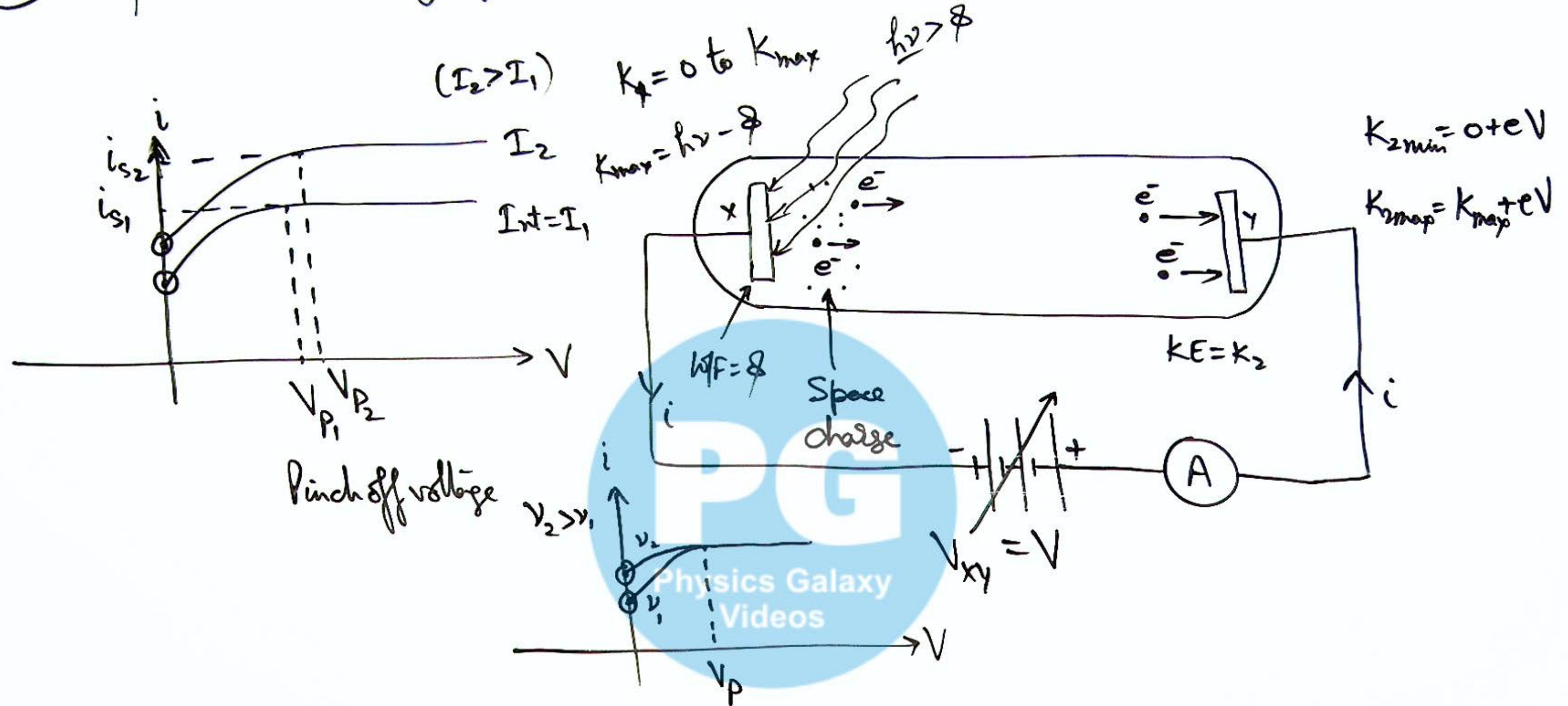
"No time lag"

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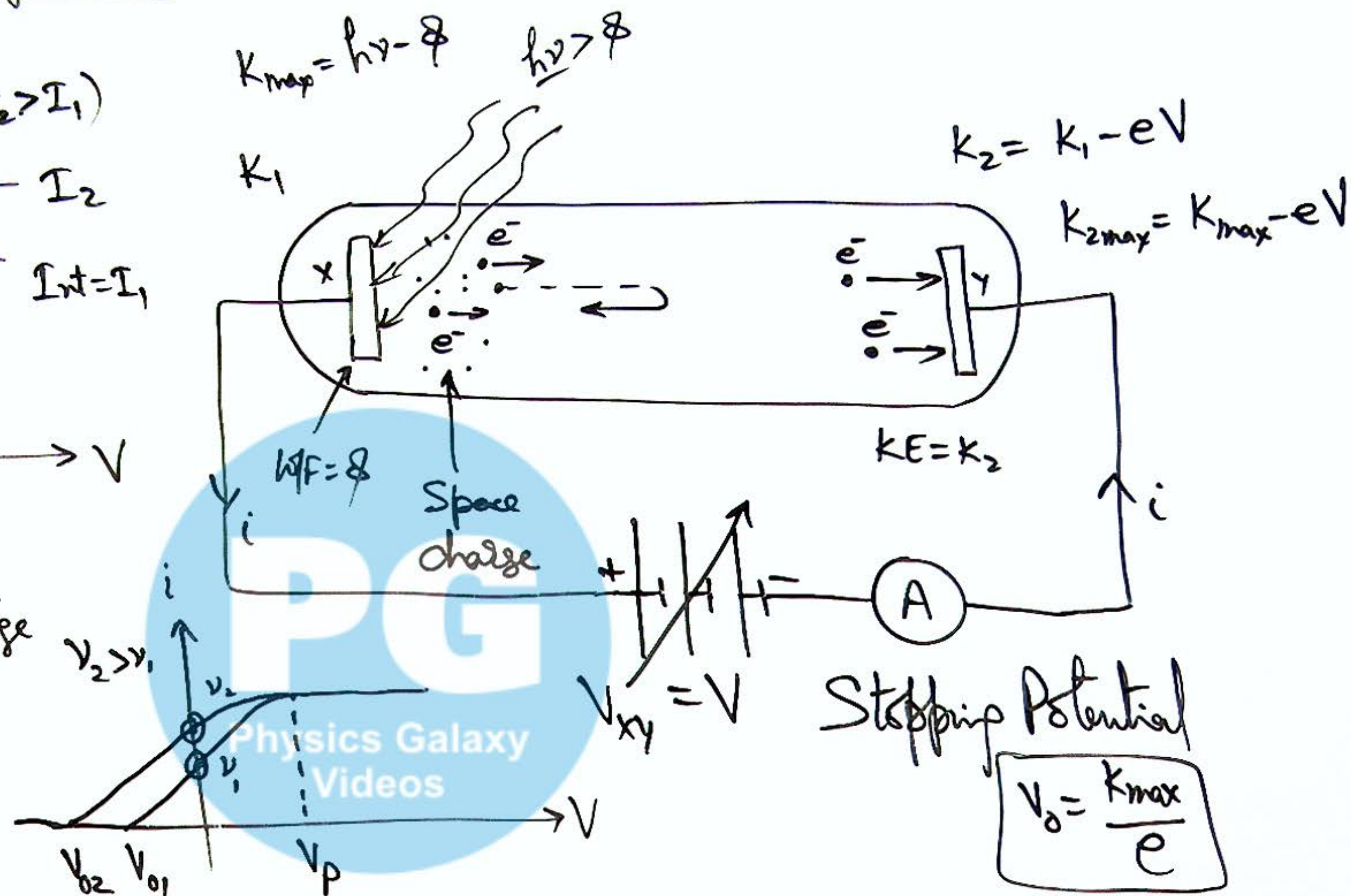
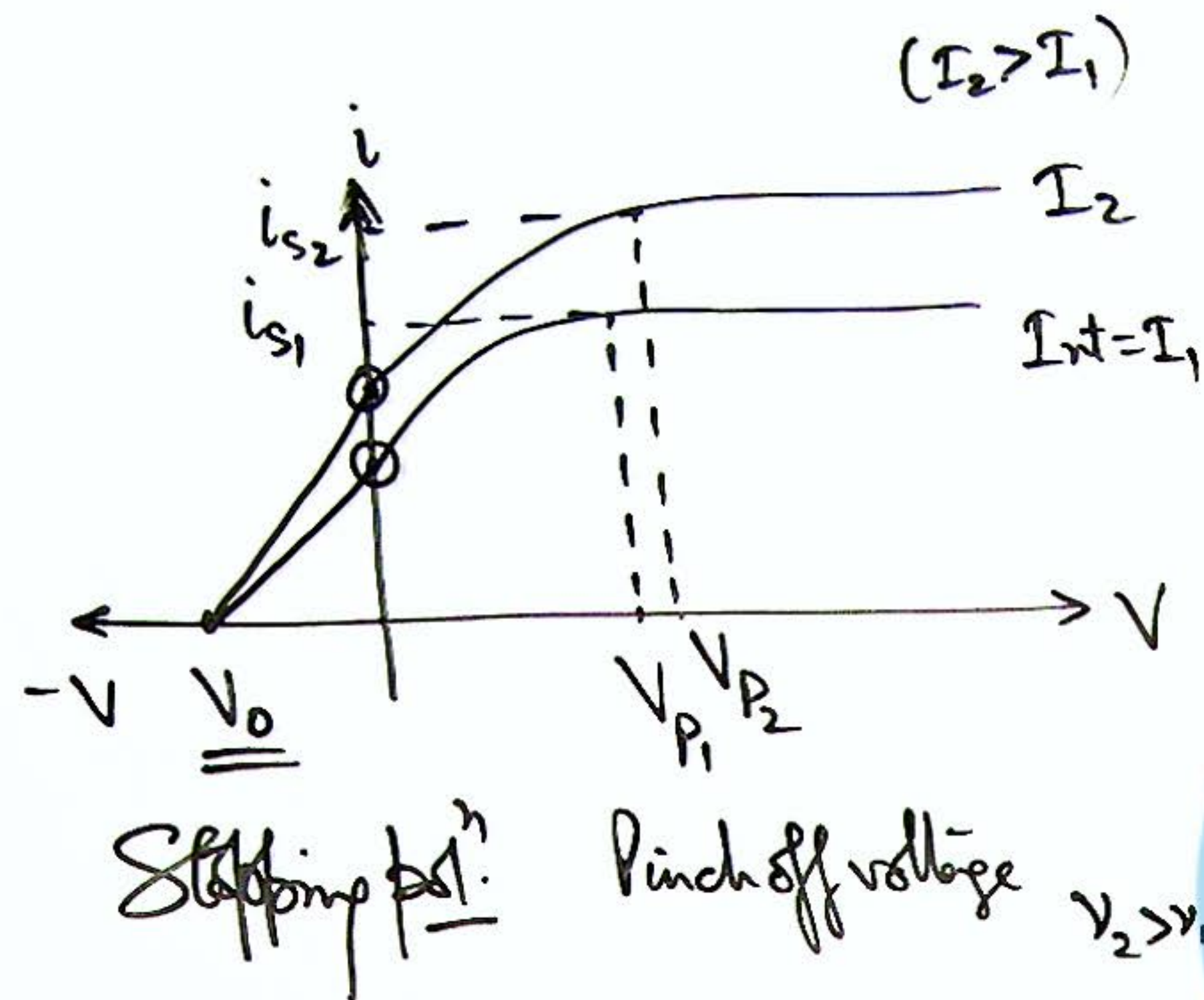
→ betwⁿ absorption of a photon
& emission of e^- .

W.D by batt on $e = eV$

⑧ Experimental Study of PEE:



⑧ Experimental Study of PEE:



⑨ Stopping Potential of a metal Surface:

Stopping potential is given as

$$KE_{\max} - eV_0 = 0$$

$$eV_0 = h(\nu - \nu_{th})$$

Einstein's PEE eqⁿ

$$\Rightarrow V_0 = \frac{\frac{1}{2}mv_{\max}^2}{e}$$

$$\Rightarrow V_0 = \frac{h\nu - \phi}{e} = \frac{h\nu - h\nu_{th}}{e}$$

$$\Rightarrow V_0 = \left(\frac{h}{e}\right)\nu - \left(\frac{h\nu_{th}}{e}\right)$$

